A Resource-Efficient Algorithm–Hardware Co-Design Toward Semi-Supervised Neurological Symptoms Prediction

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Abstract*—Recent advancements in neuroscience, coupled with the rise of deep neural networks (DNNs), have markedly enhanced the ability to predict physiological disorders. Typically, these models are trained on cloud servers using labeled datasets and subsequently deployed to devices for inference. However, this conventional cloud-based training and edge inference approach is not feasible for real-life scenarios owing to issues such as data privacy, the laborious data-labeling process, and variations in conditions resulting from electrode shifts and changes in physiological status. In response to these challenges, we present resource-efficient algorithm (REAL), a resource-efficient algorithm–hardware co-design training framework. This transformer-based framework incorporates direct feedback alignment (DFA) to optimize hardware and computational efficiency during the training process. Moreover, it utilizes semi-supervised learning (SSL) along with bespoke data augmentation techniques to reduce the dependency on extensive data labeling. Based on the proposed algorithmic model, we design and implement a dedicated hardware architecture. We tested REAL on both electroencephalogram (EEG)-based seizure prediction and electrocardiogram (ECG)-based arrhythmia classification tasks. For the EEG task, our framework enhances prediction sensitivity by an average of 7.24% and decreases the false-positive rate (FPR) by 0.04/h, demonstrating robust performance across varying levels of annotated data. With minimal labeled data, REAL reached an AUC of 0.91 and a sensitivity of 89.15%, outperforming all previous works with similar task settings. For the ECG task, REAL achieved an accuracy of 82.81% and a macro-*F*1-score of 0.827 under a 20% labeling rate, demonstrating strong generalization to a different biomedical modality. Additionally, REAL is the first hardware design targeting semi-supervised neurological symptom prediction, achieving an energy efficiency of 3.97 TOPS/W and an area efficiency of 2.39 GOPS/mm^2, both of which exceed all previous related works.

***Index Terms*—Direct feedback alignment (DFA), electroencephalogram (EEG), on-device learning, semi-supervised learning (SSL).**

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I. Introduction

As THE aging population increases, chronic diseases have become the primary cause of human mortality, imposing a substantial economic impact on society. According to data from the World Health Organization, noncommunicable chronic diseases claim the lives of 41 million people globally each year, accounting for three-quarters of global mortality [1]. The Centers for Disease Control and Prevention in the United States reports an annual expenditure of 3.7 trillion on chronic diseases [2], [3]. Therefore, how to effectively reduce the mortality rate of chronic diseases and alleviate the resulting socio-economic burden has become an urgent issue.

The Internet of Medical Things (IoMT) is considered a low-cost solution to mitigate the harm of chronic diseases. IoMT predicts the risk of chronic disease onset by analyzing physiological signals, enabling early intervention. With the development of artificial intelligence and wearable devices, a series of deep learning-based IoMT studies have emerged [4], [5]. Despite the remarkable achievements of these studies in recent years, the main computational complexity of existing systems typically lies in the cloud. Usually, after offline training, the model parameters are deployed to hardware devices for online prediction [6]. However, this reliance on cloud-based biological signal processing involves private data, which may lead to risks of data breaches. Furthermore, physiological signaling biomarkers may undergo temporal variations. These variations are often attributed to shifts in electrode positioning or transitions in the subject's physical state, which requires the device to have the ability to track these changes [7], [8]. To address data privacy and pattern change issues, adopting on-device customization learning emerges as a promising solution. However, for chronic disease symptom prediction devices, the following challenges still exist.

First, the customized calibration of devices requires a massive amount of annotated data, necessitating cumbersome data collection and re-labeling efforts [9]. This is compounded by the fact that acquiring extensive labeled physiological signal data is extremely demanding. Labeling physiological signals requires professional knowledge and experience, consuming significant time and effort, thus often resulting in high labeling costs. Additionally, the labeling of physiological signals often exhibits subjectivity and inconsistency, making it difficult to ensure the quality of labeled data. Second, on-device learning requires IoMT to support not only deep neural network (DNN) inference but also DNN training. However, the computational complexity of training for wearable devices far exceeds that required for inference, placing a substantial constraint on computational resources [10]. At the same time, compared to inference, on-chip training requires storing intermediate results of each layer, resulting in significant storage overheads and access costs.

To resolve the aforementioned issues, we propose a software–hardware co-design tailored for physiological signal processing IoMT devices, called resource-efficient algorithm (REAL). We mitigate the dependency of IoMT devices on labeled data by implementing a semi-supervised learning (SSL) approach. Additionally, by integrating the hardware-friendly training method into a transformer-based semi-supervised framework, we effectively reduce storage overheads and access costs. Furthermore, we design a specialized hardware architecture designed to accommodate the reusability and parallelism of the outlined training methodology.

The core contributions of this research include the following.

1) We propose REAL, the first transformer-based semi-supervised algorithm–hardware co-design training framework for neurological symptoms prediction.
2) The proposed framework employs hardware-friendly direct feedback alignment (DFA) to optimize computational and hardware efficiency during model training. Additionally, it utilizes SSL and custom data augmentation techniques to minimize the need for extensive data labeling.
3) A specific hardware architecture is developed to accommodate reusability and parallelism and demonstrate our proposed approach for predicting neurological symptoms. Comprehensive experimental results reveal that the presented system achieves notable effectiveness.

This article is structured into the following sections. Section II provides a survey on semi-supervised training and on-chip learning. Section III presents the model architecture, resource-efficient DFA training, and semi-supervised training algorithms, followed by the hardware design in Section IV. Section V outlines the results, and Section VI summarizes the article.

II. Related Works

A. *Semi-Supervised Algorithms for Neurological Symptoms Prediction*

SSL bridges the gap between unsupervised and supervised learning, aiming to leverage the strengths of both while mitigating their weaknesses. The main concept of SSL lies in leveraging unlabeled data as part of the training process. This approach allows models to capture broader patterns and relationships in unlabeled data, achieving performance comparable to that obtained with larger labeled datasets. In the context of neural signals, this capability is particularly valuable. Labeling such signals requires extensive manual effort and substantial time investment, as it depends heavily on expert input from medical professionals with specialized knowledge in neuroscience. Consequently, obtaining sufficiently labeled neural signals in the required quantities poses significant challenges.

Recently, numerous researchers have attempted to introduce SSL into tasks related to neurological symptoms prediction to address the degradation in performance resulting from the lack of sufficient labeled data. Abdelhameed and Bayoumi [11] proposed a semi-supervised approach leveraging transfer learning for seizure prediction. The approach utilizes an autoencoder to extract spatial features from unlabeled multichannel electroencephalogram (EEG), followed by the use of Bi-LSTM to capture temporal features, achieving spatiotemporal joint modeling. This is a representative self-supervised method capable of making predictions up to 1 h before a seizure. The reported average sensitivity is 94.6%, with a low false-positive rate (FPR) of 0.04 FP/h. However, its temporal modeling is limited to the LSTM architecture. While Bi-LSTM is suitable for modeling temporal dependencies, it faces performance bottlenecks compared to more advanced architectures such as Transformers, especially when modeling long-range dependencies. Truong et al. [12] proposed a generative adversarial network (GAN)-driven semi-supervised framework, which was tested on multiple neurological symptoms prediction datasets, with results for some patients comparable to fully supervised models. However, GAN-based semi-supervised frameworks are prone to instability during the training process, such as mode collapse, and are highly sensitive to model architecture and hyperparameter settings. Consistency regularization is one of the commonly used methods in semi-supervised training. It is based on the smoothness assumption and the clustering assumption, which posit that data points with different classifications are divided by low-density regions, whereas similar data points produce analogous outputs. Liang et al. [13] were the first to apply consistency regularization to the task of seizure prediction. Even with only a small amount of annotated data, their proposed model effectively utilized unlabeled data to optimize the decision-making boundaries for enhanced prediction performance. However, the consistency regularization-based method relies on perturbations such as Gaussian noise to enhance model robustness. Its performance is highly sensitive to the perturbation strategy, leading to significant fluctuations in performance. Additionally, the network architecture used is relatively simple, failing to fully exploit the relationships between sequential information or signals, resulting in a relatively high FPR in the model. In TIM'23, the researchers proposed a semi-supervised domain adaptation seizure prediction model (SSDA-SPM) for seizure prediction [14]. The feature alignment module in SSDA-SPM minimizes the feature distribution differences across source and target domains by measuring the discrepancies between different patients. Subsequently, the consistency regularization module strengthens the discrimination of the target subject by repositioning the decision boundary into regions of lower density. However, the residual convolutional neural network (CNN) used fails to fully exploit the correlation information between EEG signal channels, which limits the model's expressive power and predictive performance.

B. Neurological Symptoms Prediction Hardware With On-Chip Learning

Over the past decade, numerous processors supporting online learning for physiological signal processing have been proposed. Huang et al. [15] introduced an SVM-based processor that utilizes multiple multipliers in an alternating direction method for online training aimed at seizure detection. Building on this, the same team developed a processor dedicated to neural signal analysis for the prediction of seizures [16]. This processor effectively reduces hardware resource consumption for feature extraction using a low-complexity approximate energy operator. The implementation of a scaled Newton-Raphson divider extends the numerical range and reduces latency. Additionally, pointer-based matrix multiplication and LDL decomposition with rearranged computation sequences significantly lower the computational complexity of training.

Compared with classical machine learning-based methods, deep learning-driven implementations can avoid manual feature extraction, but their on-chip learning implementation requires substantial computational and storage resources. Liu et al. [17] proposed BioAIP, a physiological signal-processing chip that updates the feature vector of CNN using a template update for on-chip learning in simple physiological signal tasks. To reduce the hardware resource burden associated with backpropagation (BP) based on-chip learning, Zhao et al. [18] introduced a fully supervised on-chip learning chip, HybMED. This chip uses a reconfigurable homogeneous core and heterogeneous data flow along with a multicomputational paradigm sparse exploration design, achieving an efficiency of 1.5 TOPS/W and an area efficiency of 1.17 GOPS/mm^2.

However, the aforementioned chips typically implement fully supervised on-chip learning and are tested using publicly labeled datasets. Given that labeling physiological signals requires expert medical professionals, obtaining a large volume of high-quality labeled physiological signals is challenging in practical applications. Ronchini et al. [19] proposed an unsupervised online learning method based on spiking neural networks (SNNs) for seizure detection. Their chip, based on an analog implementation, supports only a 2×1 neuron SNN model using spike-timing-dependent plasticity (STDP) for online learning, achieving only moderate accuracy. Tsai et al. [20] proposed an on-chip training processor based on neural pattern clustering. This processor uses preexisting EEG databases for offline training to automatically extract features of neural patterns, followed by zero-shot online learning when deployed to unseen patients.

III. Proposed Algorithm Framework

Our approach integrates a hardware-friendly DFA with an SSL strategy built on a tailored Transformer architecture. In this section, we begin by outlining the model architecture utilized in our method. Subsequently, Sections III-B and III-C provide detailed explanations of the hardware-friendly algorithm and semi-supervised training strategy, respectively.

A. Network Architecture

Until recently, CNNs have maintained a dominant position in the field of biomedical research [21], [22]. Researchers have developed numerous CNN-based solutions for physiological signal-processing tasks [23], [24]. However, due to the limited kernel size of CNNs and their local receptive field characteristics, they cannot capture the long-term dependencies crucial for time-series data. In contrast, Transformers, based on self-attention mechanisms, excel in capturing long-range dependencies in signals, which is highly beneficial for handling complex correlations in physiological signals [25]. Physiological signals are nonstationary, and their complex features in the time domain often span long ranges. Traditional sequence modeling networks, such as RNNs, struggle to capture long-range dependencies due to the vanishing gradient problem. In contrast, the Transformer model, with its self-attention mechanism, directly models relationships between any time points, enabling the simultaneous analysis of global coupling properties of multifrequency rhythms (such as the α and β waves in EEG), significantly enhancing the detection of cross-window events like epileptic spikes. Furthermore, multichannel EEG contains spatial topological information about brain regions. The Transformer overcomes the limitations of fixed convolutional kernels in traditional CNNs by automatically learning interchannel functional connectivity patterns through dynamic attention weights, allowing for precise capture of interchannel correlations and effective extraction of spatiotemporal features in physiological signal processing.

The proposed network architecture is based on the vision transformer (ViT) framework, specifically designed for processing neurophysiological signals [26]. As depicted in Fig. 1, each input signal sequence (23×5120) is initially divided into patches of size 1×8, resulting in 23×640 patches. These patches are then input into a linear projection layer. Specifically, we use a 1-D CNN along the time dimension with a 1×8 convolutional kernel and a stride of 8 for embedding and mapping, ensuring that the channel information is fully preserved. Due to the segmentation and flattening of the data, the Transformer model cannot directly recognize positional information from the neurophysiological segments. To address this issue, we introduce positional embeddings, which are combined with the patches to supplement the model with positional information. Additionally, we include a learnable class token to fully represent the information within the input patches. This class embedding is typically initialized as a random vector with the same dimension as the patch embeddings. It is learnable and is used to perform global feature aggregation across all other tokens, allowing the output of this embedding to be used for classification. The processed sequence is then passed through the Transformer encoder, featuring a multihead attention mechanism with four heads and a multilayer perceptron (MLP), combined with subsequent residual connections and layer normalization. After encoding,

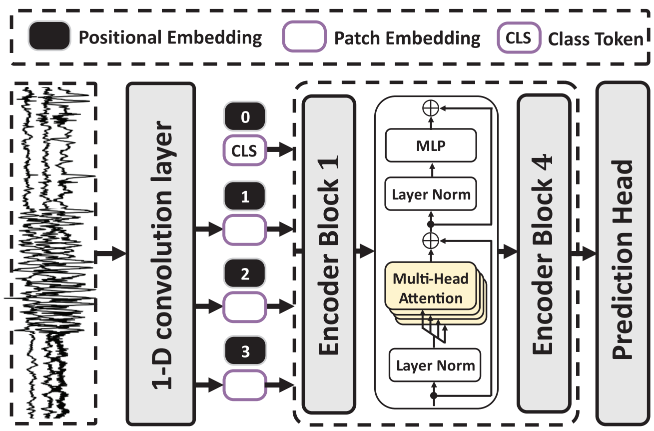


Fig. 1. Architecture of the proposed Transformer. The dimension of each input EEG signal sequence is 23×5120, which is divided into 1×8 sized blocks, resulting in 23×640 blocks. These blocks are first passed through a 1-D CNN for block embedding and mapping, utilizing a 1×8 convolutional filter and a stride of 8. Positional embeddings and learnable class tokens are introduced to provide the model with both positional and categorical information. The processed sequence is then input into the encoder block, which contains a four-head self-attention unit and an MLP, along with subsequent residual connections and layer normalization. After encoding, the high-level representation of the neurophysiological signals is passed to the MLP for final processing. The model contains a total of 60710 parameters.

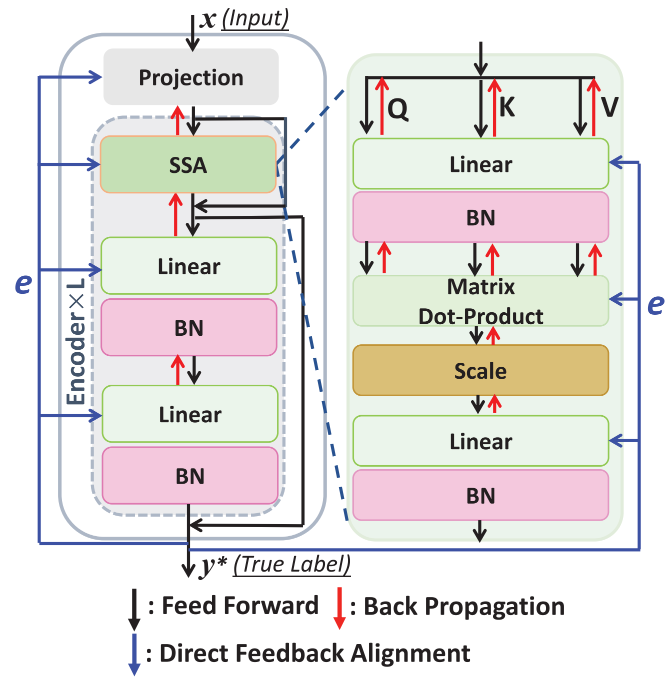


Fig. 2. Training flow with the BP and DFA algorithms in the Transformer architecture. The black arrows represent the feed-forward path, the red arrows indicate the feedback path based on backpropagation, and the blue arrows represent the feedback path based on DFA. The proposed REAL training framework is based on the DFA algorithm, aimed at addressing the weight transfer problem and providing a more hardware-friendly training algorithm.

the high-level representation of the neurophysiological signals is passed to the MLP for final processing.

B. Resource-Efficient Training

The BP algorithm is widely recognized as one of the most effective methods for training neural networks and is widely used in various deep learning models. While it demonstrates strong performance in various tasks, BP presents two nontrivial challenges for hardware implementation. First, backward locking poses a notable limitation. As illustrated in Fig. 2, BP relies on sequential, layer-by-layer error propagation during the backward pass, which restricts the feasibility of parallel processing in hardware systems. Second, in BP, buffering the input and output values of all layers is necessary during both forward and backward phases to compute weight updates, which leads to memory overhead. DFA is a potential solution to address the challenges of hardware implementation of BP.

DFA enables direct updates to all hidden layers by substituting the gradient with a random projection of the error between network predictions and targets (true labels). Unlike the sequential processing of each layer in BP, DFA's direct error propagation resolves the issue of BP locking [27]. This allows hardware implementations using DFA to update the weights of all layers simultaneously. While DFA performs effectively on fully connected networks, it struggles with training CNNs, which are widely applied in neural symptom prediction. However, Transformers can be efficiently trained based on the DFA method [28]. As shown in Fig. 2, errors generated from the predictions and true labels of the Transformer are propagated to the self-attention layers and fully connected layers. For all layers except the final linear layer, these errors are multiplied by fixed random matrices Bi of suitable dimensions. This avoids the need to store forward activation values during the training phase, effectively reducing storage overhead in hardware implementation.

C. Semi-Supervised Learning

First, we present the basic concepts of SSL applied to neurological symptom prediction and define the notation used in the proposed algorithm. For a given task, let $\mathcal{D} = \{(\boldsymbol{x}_i, \boldsymbol{y}_i)\}_{i=1}^{N_d}$ represent the labeled electrophysiological signal data, where $\boldsymbol{x}_i$ denotes the electrophysiological signal and $\boldsymbol{y}_i$ represents the corresponding label of the signal. Let $\mathcal{U} = \{\boldsymbol{u}_i\}_{i=1}^{N_u}$ represent the unlabeled samples. N_d and N_u indicate the sample counts in the sets $\mathcal{D}$ and $\mathcal{U}$, respectively. Let $f(x)$ represent the probability distribution over classes generated by the neural network f for an input x. To better focus on the training strategy, the network f was simplified to several sequential layers, where each layer computes $z_i = W_i h_{i-1} + b_i$ and $h_i = \sigma(z_i)$ using weights W_i, biases b_i, and the activation function $\sigma(\cdot)$. Here, $x = h_0$ is the input, and $f(x) = h_n$ is the final output, representing the class probability distribution. The cross-entropy loss, denoted as $H(p, q)$, measures the difference between the true class distribution p and the predicted distribution q.

Following the practices of common SSL frameworks [29], [30], [31], our proposed framework REAL integrates two core elements: consistency regularization and pseudo-labeling. Consistency regularization relies on the principles of smoothness and cluster separation, positing that instances with distinct attributes should group in low-density regions, whereas similar instances should produce consistent outputs. Specifically,

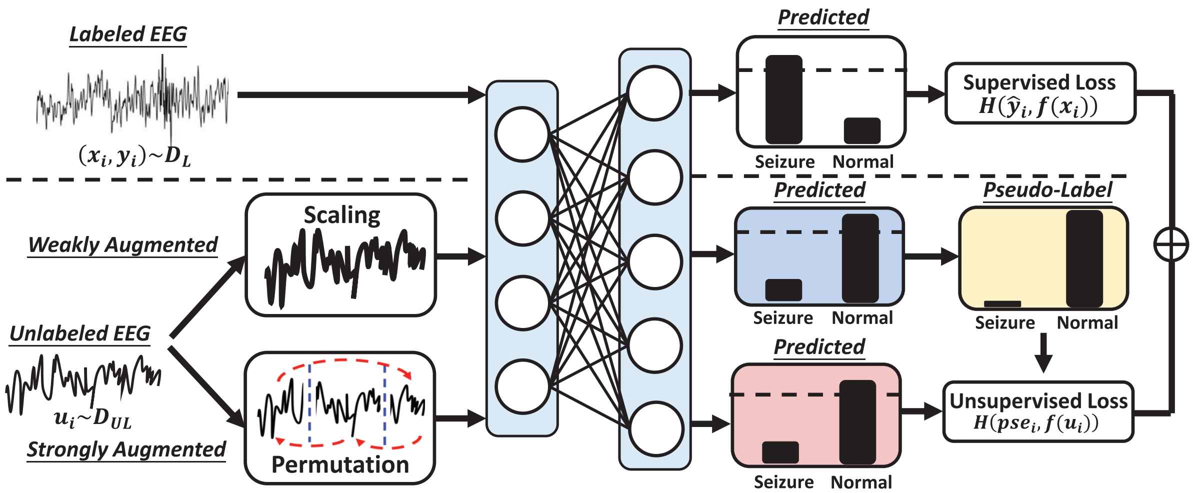


Fig. 3. Illustration of the proposed REAL semi-supervised training process. The dotted line distinguishes the supervised training flow (above) from the unsupervised one (below). Cross-entropy loss is applied to labeled data training, following standard practice. For unlabeled data, we begin by setting a threshold for the predicted probability distribution of the electrophysiological signal with weak augmentations. Samples with confidence levels exceeding the threshold are then assigned one-hot pseudo-labels. Finally, we enforce consistency regularization between the outputs of weak and strong augmentations of the samples using cross-entropy loss.

different augmentations for the same input should lead to consistent embeddings in the latent space and yield analogous predicted class distributions. On the other hand, pseudo-labeling in our framework involves generating labels for unlabeled samples based on their predicted outcomes. This technique assigns pseudo-labels to those unlabeled data points whose highest class probability surpasses a set confidence threshold [32].

As shown in Fig. 3, we use the standard cross-entropy loss as the supervised loss $\mathcal{L}_d$ for labeled data points

$$\mathcal{L}_d = \frac{1}{N_d} \sum_{i=1}^{N_d} H(y_i, f(x_i)). \tag{1}$$

For the unlabeled samples, two versions of each data point are generated by applying both weak and strong augmentations. The confidence score is determined as the highest value of the predicted probability distribution produced by the function f and is compared to a predefined threshold. If the confidence score exceeds the threshold, the cross-entropy loss is computed between the predictions corresponding to the weakly and strongly augmented samples. In mathematical terms, the unsupervised loss $\mathcal{L}_u$ is expressed as

$$\mathcal{L}_u = \frac{1}{N_u} \sum_{i=1}^{N_u} \mathbb{1}\left(\max\left(f\left(\eta\left(u_i\right)\right)\right) < \tau\right) H\left(\mathrm{pse}_i, f\left(\theta\left(u_i\right)\right)\right) \tag{2}$$

where τ represents the predefined threshold, and pseudo-label pse_i is a one-hot matrix derived from $\mathrm{argmax}(f(\eta(u_i)))$. Here, η represents the weak augmentation, which applies scaling, and θ denotes the strong augmentation, which incorporates permutation and jitter, as outlined in [33]. The complete loss function is formulated as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_d + \lambda * \mathcal{L}_u \tag{3}$$

where λ balances the contributions of the two losses.

Algorithm 1 REAL Training Flow

Input: Learning rate γ, parameters $\{W_i, b_i\}_{i=1}^n$, fixed random weight matrix $\{B_i\}_{i=1}^n$, electrophysiological signal data (x, y) or u, activation function σ.

Step 1: Loss Computation:
if labeled data:
  Compute supervised loss: $L \leftarrow H(y, f(x))$.
else:
  Generate weakly augmented data: $\eta(u)$.
  Derive pseudo-label pse_i, if $\arg\max(f(\eta(u))) > \tau$.
  Generate strongly augmented data: $\theta(u)$.
  Compute unsupervised loss: $L \leftarrow H(pse_i, f(\theta(u)))$.

Step 2: Gradient Computation with DFA:
- Compute output layer error: $\delta_n = e = \frac{\partial L}{\partial z_n}$.
- Compute hidden layer errors in parallel

$$\delta_i = B_i \cdot e \odot \sigma'(z_i), \quad \text{for } i = 1, 2, \ldots, n-1.$$

- Gradients for each layer

$$\widetilde{\nabla} W_i = \delta_i \cdot h_{i-1}^T, \quad \widetilde{\nabla} b_i = \delta_i.$$

Step 3: Parameter Update:

$$W_i \leftarrow W_i - \gamma \widetilde{\nabla} W_i, \quad b_i \leftarrow b_i - \gamma \widetilde{\nabla} b_i, \quad \text{for } i = 1, 2, \ldots, n.$$

Finally, Algorithm 1 encapsulates the core training flow of the framework, effectively integrating the principles of SSL with a streamlined and resource-efficient computational process. This concise representation outlines a single training iteration for labeled or unlabeled electrophysiological signals.

IV. Proposed Hardware Architecture

A. Overall Architecture

In Fig. 4, the proposed overall architecture is illustrated. External input data is stored in the on-chip I/O SRAM through a direct memory access (DMA) module. The SRAM consists of two blocks of 16 KB memory banks, working in a

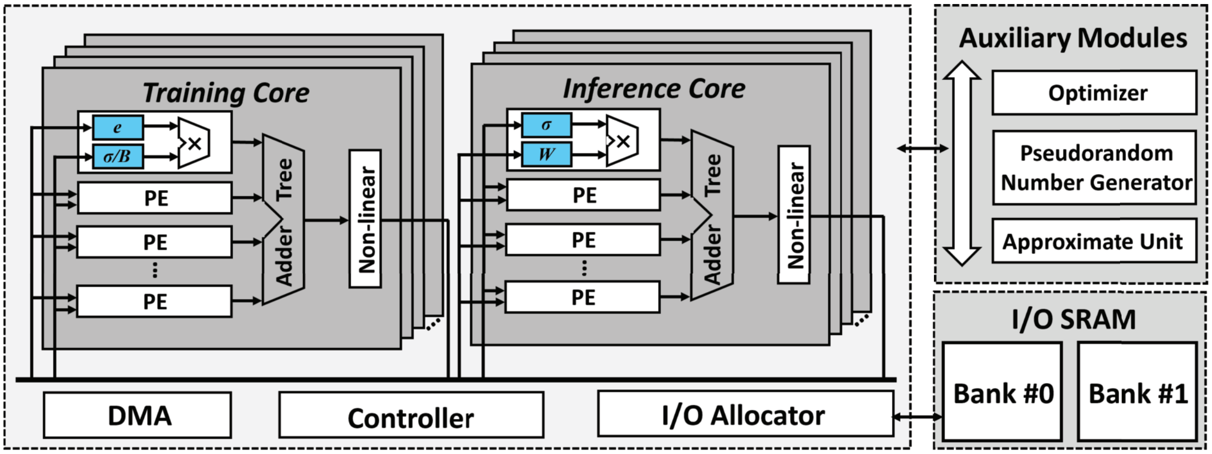


Fig. 4. Overall architecture of REAL. The architecture includes a controller, a DMA, an I/O allocator, a computation core, and auxiliary modules. The computation core contains both training and inference cores, each consisting of multiple PEs and adder trees for efficient data processing and inference computation. The auxiliary modules include an optimizer, PRNG, and approximation units, supporting functions such as optimization, random number generation, and data flow management for the core modules.

ping-pong manner. The computational cores are responsible for supporting the matrix multiplication-accumulation computations during the training and inference processes of the proposed high-efficiency semi-supervised framework. The proposed architecture consists of eight computation cores, where each core contains processing elements (PE) and an adder tree for multiplication-accumulation computations. Each computing core contains 16 PEs/MACs, totaling 128 PEs/MACs. The eight computing cores can be configured with different data flows to support both training and inference. The computed results are then processed through the nonlinear layer to generate the output of each layer. The numerical precision for computations is int-8.

To fully support the on-chip computation of the model, the auxiliary modules include additional modules for on-chip data scheduling and computations other than matrix multiplication accumulation. The pseudorandom number generator (PRNG) module generates random numbers of specific patterns on-chip by loading random number seeds. The I/O allocator is responsible for data scheduling, allocating data to computational cores based on different computation paradigms. Fig. 5 (top) shows the data input rearrangement logic and (bottom) the data output rearrangement logic. The data input section of each computing core comprises 16 source ports linked to SRAM or PRNG modules, and 16 destination ports connected to PEs. Each source port accesses a designated address (e.g., i_0) in the SRAM or PRNG module, and the data is guided to any PE for further computation based on an enable signal (e.g., ie_0) decoded from the instruction. The data output rearrangement section comprises 16 source ports linked to the PEs and 16 destination ports interfacing with the data SRAM. These source ports receive computational outputs (e.g., o_0) from the PEs, and each bit of the output paired with an enable signal (e.g., oe_0) that specifies which bits are written to the corresponding SRAM address. The I/O allocator can be flexibly configured to support the requirements of different computational patterns. The approximate unit contains LUTs for approximate operations of exponential and division, used for computations like Softmax. The optimizer module supports the Adam optimizer, which has shown good performance for DFA.

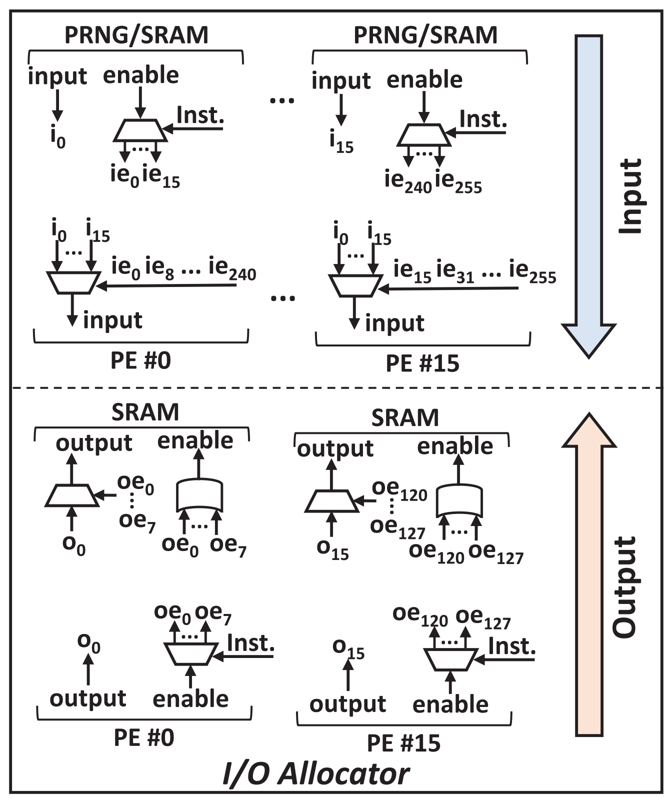


Fig. 5. Detailed implementation of the I/O Allocator. Data input rearrangement logic (top) and data output rearrangement logic (bottom). During data input, each computation core has 16 source ports connected to either the SRAM or PRNG module. During data output, the SRAM receives the outputs generated by the 16 PE source ports.

B. Reusability

The proposed semi-supervised training framework incorporates various operators. To improve the energy efficiency of the proposed hardware architecture, we explore the reusability of the proposed framework, maximizing the reuse of hardware resources. First, the proposed framework requires the use of random numbers in multiple places, including the DFA feedback random matrix B, the scaling operation for semi-supervised weak data augmentation, and the permutation operation for strong data augmentation. Therefore, we implemented an on-chip PRNG module based on the Xoroshiro128+ unit [34], capable of generating various modes of random numbers to meet the requirements of different operators. Second, the proposed framework supports both training and inference. We specifically designed a combination of reusable computational cores and heterogeneous data paths, supported by the I/O allocator to facilitate data scheduling for PRNG and I/O memory.

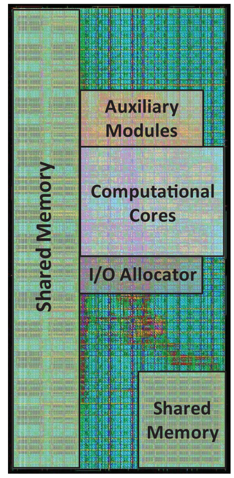
C. Parallelism

As mentioned in the previous section, DFA has the advantage of parallel computation compared to the BP algorithm. Its inference and training for each layer can be decomposed and computed in parallel. Therefore, we explore specialized strategies to leverage the parallelism of DFA. Through flexible programmable design and data scheduling strategies, the proposed hardware architecture can support the dynamic allocation of computational cores. The allocation of workload to each core is determined based on the required number of computing units for forward inference and DFA training in each layer. In other words, computational cores can be dynamically configured as either training cores or inference cores. For training cores, processing units typically handle the MAC operations of errors e with activations σ or with the random matrix B in training mode. For inference cores, processing units typically handle the MAC computations of activations σ with forward weights W. The proposed parallel exploration strategy effectively reduces processing latency while increasing the utilization of computational cores.

V. Experiments and Results

A. Experimental Settings

We evaluated the proposed method using a seizure prediction task. For a fair comparison, we utilized the publicly available CHB-MIT dataset [37], which includes 23 scalp EEG recordings collected at a 256-Hz sampling rate from 22 patients. Among these recordings, 15 were obtained using 23 fixed electrodes. Additionally, since other studies using semi-supervised methods for epilepsy prediction typically use data from a maximum of 13 patients, we excluded Chb04 and Chb05 records. Seizure prediction horizon (SPH) and preictal interval length (PIL) are two crucial parameters used to analyze seizure prediction data. SPH is defined as the duration from the onset of a seizure to the prediction warning. PIL represents the length of the prediction warning. In this study, SPH and PIL are configured to 5 and 30 min, respectively. To address practical challenges such as signal noise, electrode degradation, and patient movement, the signals used in our study were preprocessed to remove common artifacts. This ensures cleaner input while preserving physiological variability. Our deep learning model is designed to learn robust representations that account for the inherent fluctuations and noise typically present in physiological data. In the CHB-MIT dataset, seizure episodes are often separated by extended seizure-free intervals, for example, subject 10 exhibits a 16.1-h gap between the first and second recorded seizures. These periods naturally include various real-world disturbances, which are implicitly captured during training, enhancing the model's resilience in practical applications.



	Specifications
Technology	40nm CMOS
Die Area	1.50 mm^2
SRAM	32 KB
Supply	1 V
Frequency	1 MHz
Power	0.955 mW
Energy Efficiency	3.97 TOPS/W
Area Efficiency	2.39 GOPS/mm^2

Fig. 6. Floorplan and performance summary of the proposed hardware.

Thirteen patients with fixed acquisition configurations were selected for this study. The results reported in the article are based on five independent random experiments. For each experiment, we randomly split the preictal and interictal data for each patient. The data of each patient were randomly split into 80% for training and 20% for testing. Both types of samples had a sliding window of 20 s, with preictal samples having a 75% overlapping sampling window and interictal samples having nonoverlapping sliding sampling windows to mitigate the decrease in accuracy due to data imbalance. We employ stochastic gradient descent (SGD) as the optimizer with weight decay set to 5×10^{-4} and use a cosine scheduler. The initial learning rate was set to 3×10^{-2}, batch size to 64, and a predefined threshold of 0.95. Performance was assessed using sensitivity, FPR, area under the curve (AUC), and $F1$-score.

The proposed processor is implemented in Verilog and synthesized using the Synopsys design compiler with the TSMC 40-nm library. We evaluated the SRAM performance using the TSMC memory compiler. Fig. 6 shows the floorplan and performance summary of the proposed processor, while Fig. 7 presents the area and power breakdown results. Using a 40-nm CMOS process, the proposed processor has a die area of 1.5 mm^2, which includes 32 KB of SRAM. At a frequency of 1 MHz, it achieves a power consumption of 0.955 mW, a peak energy efficiency of 3.97 TOPS/W, and a maximum area efficiency of 2.39 GOPS/mm^2. In addition, we calculated the PE utilization rates under different computational patterns (patch embedding, attention, MLP, and DFA),

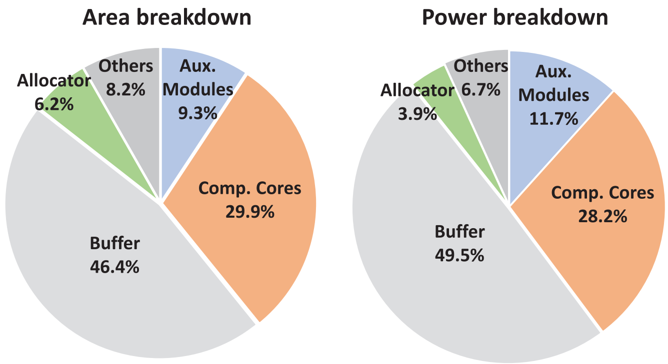


Fig. 7. Area and power breakdown of the proposed processor.

including instruction configuration, data loading, computation, and result output. The patch embedding achieved an average PE utilization rate of 93.03%, attention achieved 86.96%, MLP achieved 83.33%, and DFA achieved 68.97%.

In addition, we evaluate the performance of electrocardiogram (ECG)-based arrhythmia detection using the MIT-BIH dataset, which contains recordings from 47 subjects. The dataset consists of 48 half-hour excerpts of two-channel ambulatory ECG recordings. Among these, 23 recordings were randomly selected from a pool of approximately 4000 24-h ambulatory ECG samples collected from both inpatients and outpatients at Beth Israel Hospital in Boston, with an in-patient to an out-patient distribution ratio of approximately 3:2. The remaining 25 recordings include less common but clinically significant arrhythmias that are underrepresented in randomly selected samples. All recordings were digitized with 11-bit resolution over a 10-mV range at a sampling rate of 360 Hz per channel. We excluded patients with paced heartbeats [37].

Following the Association for the Advancement of Medical Instrumentation (AAMI) standard, we categorized the beats into five classes: nonectopic (N), ventricular ectopic (VEB), supraventricular ectopic (SVEB), fusion (F), and unknown (Q). We extracted 90082 N samples, 7008 VEB samples, 2781 SVEB samples, 802 F samples, and 15 Q samples from the dataset. To address the class imbalance, we randomly selected 800 samples each from the N, VEB, SVEB, and F categories. Q samples were excluded due to their insufficient quantity for effective splitting into labeled and unlabeled sets. The resulting 3200 samples were randomly divided into training and testing sets using an 80:20 split. The network architecture for ECG classification is similar to that illustrated in Fig. 1, consisting of an embedding layer, three encoder layers with two attention heads each, and a final MLP for prediction. The input shape is adapted for ECG signals as 2×288, with an embedding dimension of 32 and a hidden layer size of 64. The ECG signal is segmented into nonoverlapping frames using a filter size of 1×36 and transformed into a sequence of patch embeddings via a 1-D convolutional layer. This results in a lightweight model with a total of 138660 trainable parameters.

Fig. 8 summarizes our training and testing procedure, illustrating the data flow of labeled and unlabeled samples, the application of weak and strong augmentations, the computation of supervised and unsupervised losses, and the gradient update process using the DFA algorithm.

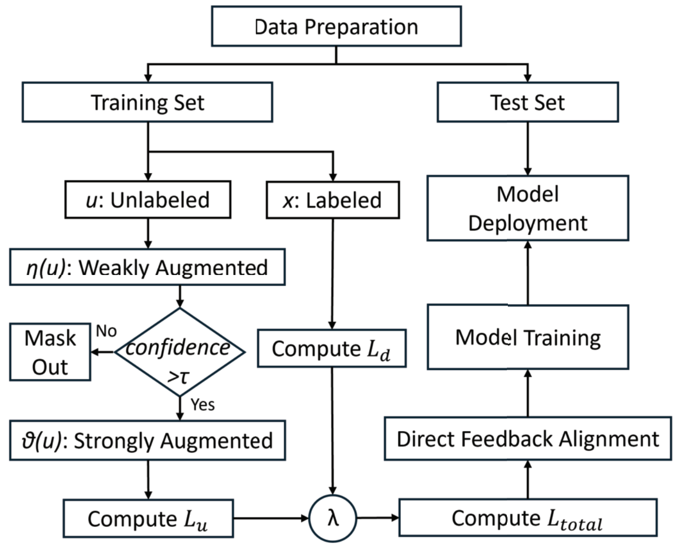


Fig. 8. Flowchart of the proposed REAL semi-supervised training and testing process.

TABLE I
PERFORMANCE ON THE CHB-MIT DATASET UNDER DIFFERENT LABEL RATES WITH AND WITHOUT UNLABELED DATA

Methods	Label Rate	Sensitivity (%)	FPR (h^{-1})	AUC	F1 Score
Only Labeled Data	**1%**	60.08	0.39	0.66	0.18
	5%	70.92	0.22	0.81	0.36
	10%	77.38	0.17	0.86	0.44
	20%	79.33	0.16	0.88	0.46
	Avg.	**71.93**	**0.24**	**0.80**	**0.36**
Labeled and Unlabeled Data	**1%**	68.84	0.30	0.72	0.29
	5%	78.08	0.19	0.85	0.41
	10%	80.62	0.15	0.89	0.47
	20%	89.15	0.15	0.91	0.58
	Avg.	**79.17**	**0.20**	**0.84**	**0.44**

B. Performance Results

To examine the performance of REAL framework, 1%, 5%, 10%, and 20% of the training samples were selected as labeled data. Table I presents the performance across various labeling rates for CHB-MIT dataset. As opposed to fully supervised learning with only labeled data, the proposed framework combines labeled and unlabeled data, achieving improvements of 7.24% in average sensitivity, 0.04/h in FPR, 0.04 in AUC, and 0.08 in the $F1$-score across different labeling rates. These results validate the efficacy of our semi-supervised framework. Furthermore, with a reduction in labeling rate, a notable improvement is observed through the utilization of unlabeled data. Specifically, the results show that REAL achieves the most significant performance improvement under extremely low annotation (1%), with an 8.76% increase in sensitivity and a 0.11 improvement in the $F1$-score. This is particularly valuable for real-world clinical applications where

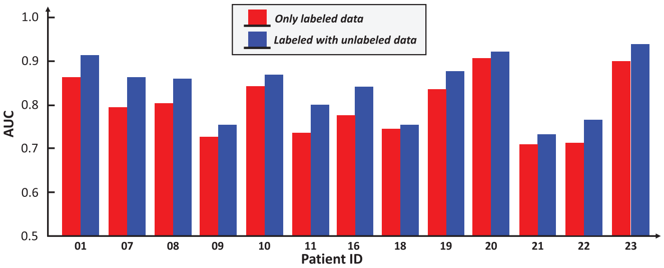


Fig. 9. Comparison of AUC between using only labeled data and combining labeled and unlabeled data.

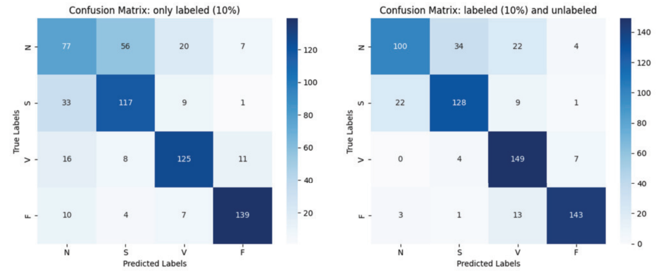


Fig. 10. Confusion matrix of only labeled data and labeled and unlabeled data under 10% label rate, respectively.

TABLE II
PERFORMANCE ON THE MIT-BIH DATASET UNDER DIFFERENT LABEL RATES WITH AND WITHOUT UNLABELED DATA

Methods	Label Rate	Accuracy (%)	F1 Score
Only Labeled Data	**1%**	56.87	0.557
	5%	68.12	0.673
	10%	71.56	0.713
	20%	76.56	0.759
	Avg.	**68.28**	**0.676**
Labeled and Unlabeled Data	**1%**	67.50	0.664
	5%	75.47	0.745
	10%	81.25	0.809
	20%	82.81	0.827
	Avg.	**76.76**	**0.761**

obtaining large amounts of labeled data is challenging. In the 5%–10% annotation range, REAL continues to demonstrate stable improvements, indicating that incorporating unlabeled data can effectively enhance model discriminability even with moderate amounts of labeled data. At a 20% annotation rate, despite already strong baseline performance, the inclusion of unlabeled data further boosts sensitivity to nearly 90% and increases the $F1$-score by 0.12, suggesting that the semi-supervised mechanism of REAL still provides marginal gains in high-performance regimes. Additionally, the FPR consistently decreases, with a notable drop of 0.09/h at the 1% annotation rate, indicating that the REAL framework not only improves detection capability but also mitigates the risk of over-activation.

Fig. 9 illustrates the comparison of average AUC between fully supervised learning using only labeled data and combining labeled and unlabeled data for each patient. The results indicate a significant enhancement in performance achieved by the proposed semi-supervised framework for each patient. The left side of Fig. 6 presents the layout of the proposed accelerator, while the table on the right summarizes the synthesis parameters and results. The proposed accelerator occupies an area of 1.5 mm^2 and consumes 955 μW of power at a working frequency of 1 MHz. Additionally, it exhibits a maximum energy efficiency of 3.97 TOPS/W and a maximum area efficiency of 2.39 GOPS/mm^2.

To further assess the applicability of the REAL framework beyond seizure prediction, we evaluate its performance on the MIT-BIH arrhythmia dataset. As shown in Table II, experiments are conducted with 1% (24 samples), 5% (128), 10% (256), and 20% (512) labeled data, maintaining class balance across the four ECG categories. The 1% split is slightly rounded from its theoretical value to ensure integer sample counts.

Across all labeling rates, incorporating unlabeled data consistently improves both accuracy and macro-$F1$-score compared to training with labeled data alone. On average, the REAL framework achieves an 8.48% increase in accuracy and a 0.08 improvement in the $F1$-score. The most pronounced gain is observed at the 1% label rate, where accuracy improves by 10.63%, demonstrating the framework's ability to perform well even with minimal supervision. From the 1%–5% interval, both labeled-only and semi-supervised settings exhibit notable improvements in accuracy, indicating that even a small increase in labeled data can substantially enhance model discriminability. At the 10% level, the model reaches 81.25% accuracy and 0.809 $F1$-score, reflecting stable and effective use of unlabeled data. Fig. 10 presents confusion matrices under the 10% condition, illustrating more accurate predictions across all classes when unlabeled data are leveraged.

Under the 20% labeling setting, REAL continues to offer marginal gains, achieving the highest recorded $F1$-score of 0.827 alongside an accuracy of 82.81%. These results indicate that REAL generalizes effectively to ECG-based classification tasks and maintains reliable performance across varying levels of supervision, without requiring modifications to its model architecture.

C. Comparison With the State-of-the-Art Works

Table III presents a comparison between the proposed semi-supervised training framework and other state-of-the-art semi-supervised seizure prediction methods. To achieve a fair comparison, we utilized the same dataset and partitioning of SPH and PIL as the references. Liang et al. [35] proposed a semi-supervised domain adaptation framework that adapts multiple source-specific models to a target patient without accessing source data. Using limited labeled samples and knowledge distillation, the method achieved 88.6% sensitivity and 0.856 AUC. Qi et al. [36] proposed a semi-supervised approach based on deep pairwise representation alignment, utilizing only 25 labeled samples per class and achieving 87.65%

TABLE III
COMPARISON OF PERFORMANCE WITH STATE-OF-THE-ART METHODS

Ref.	Dataset	Patient	Method	SPH	PIL	AUC	Sensitivity	FPR
Abdelhameed et al. [11]	CHB-MIT	12	Transfer Learning	/	/	/	94.60%	0.04
Truong et al. [12]	CHB-MIT	13	Generative Adversarial Network	5 min	30 min	0.78	/	/
Liang et al. [13]	CHB-MIT	11	Consistency Regularization	5 min	30 min	/	82.60%	0.21
Liang et al. [14]	CHB-MIT	13	Feature Alignment	5 min	30 min	0.85	88.80%	0.182
Liang et al. [35]	CHB-MIT	13	Domain Adaptation + Semi-supervised	5 min	30 min	0.86	88.60%	0.182
Qi et al. [36]	CHB-MIT	13	Data Augmentation + Semi-supervised	5 min	30 min	/	87.65%	/
This Work	**CHB-MIT**	**13**	**Consistency Regularization + Pseudo-Labeling**	**5 min**	**30 min**	**0.91**	**89.15%**	**0.155**

TABLE IV
COMPARISON OF VARIOUS WORKS IN TERMS OF TECHNOLOGY, PERFORMANCE, AND EFFICIENCY

	This Work	JSSC'19 [15]	JSCC'22 [16]	BioAIP [17]	TBioCAS'23 [38]	ISSCC'23 [20]	TBioCAS'24
Technology	**40 nm**	40 nm	40 nm	65 nm	55 nm	40 nm	40 nm
Die Area	**1.5 mm2**	2.55 mm2	1.96 mm2	1.74 mm2	5.06 mm2	1.56 mm2	1.53 mm2
Clock Frequency	**1 MHz**	130 kHz	6 MHz	500 k/ 1 M/ 2.5 MHz	300 kHz	512 Hz	500 kHz
Classifier	**Transformer**	SVM	SVM	CNN	CNN	CNN	CNN
Online Training	**REAL**	ADMM	/	Template update	/	NPC	DFA
Supervised Type	**Semi-supervised**	Fully-supervised	/	Fully-supervised	/	Unsupervised	Fully-supervised
Tasks	**Seizure Prediction**	Seizure Detection	Seizure Prediction	Seizure Detection	Seizure Prediction	Seizure Detection	Seizure Prediction
SRAM	**32 KB**	>35.8 KB	5 KB	73 KB	160 KB	>120 KB	32 KB
Parameters	**60710**	/	/	14117	10944	23104	10912
Power (μW)	**955***	1900	2310	30	142.9	1813.12	1200
Max Energy Efficiency (TOPS/W)	**3.97**	/	/	1.07	2.33	/	1.58
Max Area Efficiency (GOPS/mm2)	**2.39**	/	/	0.018	0.066	/	1.17

* The power consumption is derived from simulation results provided by the synthesis tool.

sensitivity. Liang et al. [14] proposed a semi-supervised algorithm based on feature alignment and consistency regularization for seizure prediction tasks, achieving an AUC of 0.85, a sensitivity of 88.8%, and an FPR of 0.182 on 13 patients from the CHB-MIT dataset. Liang et al. [13] employed a consistency regularization-based semi-supervised algorithm, reporting a sensitivity and FPR of 82.6% and 0.21, respectively. Truong et al. [12] utilized SSL based on GANs, achieving an AUC of 0.78. Abdelhameed et al. [11] employed a transfer learning-based semi-supervised method, achieving a sensitivity of up to 94.6% and an FPR of 0.04. However, due to the unclear SPH and PIL, its practical application value cannot be determined. This article first presents the hardware-aware semi-supervised seizure prediction method and conducts hardware design based on the proposed REAL method. Concurrently, our proposed method achieved an AUC of 0.91, a sensitivity of 89.15%, and an FPR of 0.155 at a labeling rate of 20%, surpassing all previous relevant works in these metrics.

Table IV illustrates a comparison between the proposed hardware implementation and existing relevant works. Hsieh et al. [16] and Zhao et al. [38] proposed seizure prediction chips based on the SVM and CNN, respectively. However, these works lack on-chip learning capability, making it challenging to deploy them in real-world scenarios. Huang et al. [15] introduced an SVM-based processor for EEG-based epileptic seizure detection, supporting on-chip learning based on the alternating direction method of multipliers (ADMM). However, it still consumes 1.9 mW of power at a working frequency of 130 kHz. BioAIP achieves high inference accuracy with as low as 30 μW of power consumption. Nevertheless, its on-chip training method is very simple, based only on template updates for updating feature vectors, limiting its applicability to simple online learning tasks. Tsai et al. [20] presented an on-chip training processor based on the neural pattern cluster (NPC), enabling zero-shot online retraining. However, this processor still consumes 1813.12 μW of power when processing the CHB-MIT dataset at 512 Hz. Zhao et al. [18] proposed a seizure prediction processor based on DFA, achieving a power consumption of 1.2 mW, energy efficiency of 1.58 TOPS/W, and area efficiency of 1.17 GOPS/mm^2. However, its fully supervised online training approach imposes limitations on its usage. We propose the REAL framework, which combines Transformer-based SSL with hardware optimization for the first time, using DFA-based online training. The proposed processor consumes 955 μW of power at a clock frequency of 1 MHz. The inherent parallelism of DFA allows for the decomposition and parallel computation of each layer during both training and inference. To leverage this, we design a reusable and parallel-supporting hardware architecture. By combining reusable computation cores with heterogeneous data paths and enhancing data scheduling through an I/O allocator, our hardware dynamically configures computation cores: the training core handles error e, activation σ, and the MAC operations of the random matrix B, while the inference core processes the MAC operations between the forward weights W and activation σ. This design fully exploits DFA's parallelism, significantly reducing processing latency, while SSL minimizes reliance on labeled data and improves training efficiency. Additionally, we integrate an on-chip PRNG based on the Xoroshiro128+ unit, which supports DFA's random feedback matrix B and semi-supervised data augmentation requirements. This avoids the overhead of external random number generation, further optimizing resource utilization. These innovations directly contribute to an outstanding energy efficiency of 3.97 TOPS/W and an area efficiency of 2.39 GOPS/mm^2, demonstrating a perfect balance between compact design and high performance.

VI. Conclusion

In this study, we present a software–hardware co-design tailored for physiological signal-processing IoMT devices, called REAL, for neurological symptom prediction. The REAL framework is implemented by a synergistic algorithm–hardware co-design. By incorporating DFA for optimized training efficiency and employing SSL along with specific data augmentation techniques, REAL optimizes hardware and computational efficiency and reduces the reliance on vast quantities of labeled data during on-chip training.

Our experimental results, specifically in the context of seizure prediction using EEG data, underscore the effectiveness of the REAL framework. REAL demonstrates an impressive increase in prediction sensitivity by an average of 7.24% and a reduction in the FPR by 0.04/h, indicating robust performance across various levels of labeled data availability. Remarkably, with a limited amount of labeled data, REAL achieved an AUC of 0.91 and a sensitivity of 89.15, surpassing all prior works with similar task settings. Moreover, the REAL framework is pioneering in its hardware design tailored for semi-supervised neurological symptom prediction. The dedicated hardware architecture achieves a notable energy efficiency of 3.97 TOPS/W and area efficiency of 2.39 GOPS/mm^2, both of which surpass the metrics of all previous related works.

References

[1] *Noncommunicable Diseases Progress Monitor*, World Health Org., Geneva, Switzerland, 2022.

[2] *Centers for Disease Control and Prevention*, CDC, Atlanta, GA, USA, 2020.

[3] S. Zhao, D. Wu, J. Yang, and M. Sawan, "A resource-efficient and data-restricted training method towards neurological symptoms prediction," in *Proc. IEEE Biomed. Circuits Syst. Conf. (BioCAS)*, Oct. 2022, pp. 615–619.

[4] J. Mathew, N. Sivakumaran, and P. A. Karthick, "Variational mode decomposition-based moment fusion for the detection of seizure types from the scalp EEG measurements," *IEEE Trans. Instrum. Meas.*, vol. 72, 2023, Art. no. 4012212.

[5] J. Yang and M. Sawan, "From seizure detection to smart and fully embedded seizure prediction engine: A review," *IEEE Trans. Biomed. Circuits Syst.*, vol. 14, no. 5, pp. 1008–1023, Oct. 2020.

[6] P. Busia et al., "Reducing false alarms in wearable seizure detection with EEGformer: A compact transformer model for MCUs," *IEEE Trans. Biomed. Circuits Syst.*, vol. 18, no. 3, pp. 608–621, Jun. 2024.

[7] L. G. Rocha et al., "Real-time HR estimation from wrist PPG using binary LSTMs," in *Proc. IEEE Biomed. Circuits Syst. Conf. (BioCAS)*, Oct. 2019, pp. 1–4.

[8] M. Salem, S. Taheri, and J. Yuan, "ECG arrhythmia classification using transfer learning from 2-dimensional deep CNN features," in *Proc. IEEE Biomed. Circuits Syst. Conf. (BioCAS)*, Oct. 2018, pp. 1–4.

[9] J. Hu, C. S. Leow, S. Tao, W. L. Goh, and Y. Gao, "Supervised contrastive learning framework and hardware implementation of learned ResNet for real-time respiratory sound classification," *IEEE Trans. Biomed. Circuits Syst.*, vol. 19, no. 1, pp. 185–195, Feb. 2025.

[10] S. Zhao, C. Fang, J. Yang, and M. Sawan, "Emerging energy-efficient biosignal-dedicated circuit techniques: A tutorial brief," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 6, pp. 2592–2597, Jun. 2022.

[11] A. M. Abdelhameed and M. Bayoumi, "Semi-supervised deep learning system for epileptic seizures onset prediction," in *Proc. 17th IEEE Int. Conf. Mach. Learn. Appl. (ICMLA)*, Dec. 2018, pp. 1186–1191.

[12] N. D. Truong, L. Zhou, and O. Kavehei, "Semi-supervised seizure prediction with generative adversarial networks," in *Proc. 41st Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2019, pp. 2369–2372.

[13] D. Liang et al., "Semisupervised seizure prediction in scalp EEG using consistency regularization," *J. Healthcare Eng.*, vol. 2022, Jan. 2022, Art. no. 1573076.

[14] D. Liang et al., "Semi-supervised domain-adaptive seizure prediction via feature alignment and consistency regularization," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–12, 2023.

[15] S.-A. Huang, K.-C. Chang, H.-H. Liou, and C.-H. Yang, "A 1.9-mW SVM processor with on-chip active learning for epileptic seizure control," *IEEE J. Solid-State Circuits*, vol. 55, no. 2, pp. 452–464, Feb. 2020.

[16] Y.-Y. Hsieh, Y.-C. Lin, and C.-H. Yang, "A 96.2-nJ/class neural signal processor with adaptable intelligence for seizure prediction," *IEEE J. Solid-State Circuits*, vol. 58, no. 1, pp. 167–176, Jan. 2023.

[17] J. Liu et al., "An ultra-low power reconfigurable biomedical AI processor with adaptive learning for versatile wearable intelligent health monitoring," *IEEE Trans. Biomed. Circuits Syst.*, vol. 17, no. 5, pp. 952–967, May 2023.

[18] S. Zhao, C. Wang, C. Fang, F. Tian, J. Yang, and M. Sawan, "HybMED: A hybrid neural network training processor with multi-sparsity exploitation for Internet of Medical Things," *IEEE Trans. Biomed. Circuits Syst.*, vol. 18, no. 5, pp. 1178–1189, Oct. 2024.

[19] M. Ronchini et al., "A CMOS-based neuromorphic device for seizure detection from LFP signals," *J. Phys. D, Appl. Phys.*, vol. 55, no. 1, Jan. 2022, Art. no. 014001.

[20] C.-W. Tsai et al., "SciCNN: A 0-shot-retraining patient-independent epilepsy-tracking SoC," in *Proc. IEEE Int. Solid- State Circuits Conf. (ISSCC)*, Feb. 2023, pp. 488–490.

[21] Y. Xu, J. Yang, S. Zhao, H. Wu, and M. Sawan, "An end-to-end deep learning approach for epileptic seizure prediction," in *Proc. 2nd IEEE Int. Conf. Artif. Intell. Circuits Syst. (AICAS)*, Aug. 2020, pp. 266–270.

[22] Y. Wei et al., "A review of algorithm & hardware design for AI-based biomedical applications," *IEEE Trans. Biomed. Circuits Syst.*, vol. 14, no. 2, pp. 145–163, Apr. 2020.

[23] V. Lawhern, A. Solon, N. Waytowich, S. M. Gordon, C. Hung, and B. J. Lance, "EEGNet: A compact convolutional neural network for EEG-based brain–computer interfaces," *J. Neural Eng.*, vol. 15, no. 5, p. 056013, 2018.

[24] S. Lee, Y. Hung, Y. Chang, C. Lin, and G. Shieh, "RISC-V CNN coprocessor for real-time epilepsy detection in wearable application," *IEEE Trans. Biomed. Circuits Syst.*, vol. 15, no. 4, pp. 679–691, Aug. 2021.

[25] K. Han et al., "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, Jan. 2023.

[26] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[27] D. Han, J. Lee, J. Lee, and H.-J. Yoo, "A 1.32 TOPS/W energy efficient deep neural network learning processor with direct feedback alignment based heterogeneous core architecture," in *Proc. Symp. VLSI Circuits*, Jun. 2019, pp. C304–C305.

[28] J. Launay, I. Poli, F. Boniface, and F. Krzakala, "Direct feedback alignment scales to modern deep learning tasks and architectures," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 9346–9360.

[29] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2020, pp. 596–608.

[30] D. Berthelot et al., "ReMixMatch: Semi-supervised learning with distribution alignment and augmentation anchoring," 2019, *arXiv:1911.09785*.

[31] D. Berthelot, N. Carlini, I. Goodfellow, N. Papernot, A. Oliver, and C. A. Raffel, "Mixmatch: A holistic approach to semi-supervised learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.

[32] D.-H. Lee et al., "Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2013, vol. 3, no. 2, p. 896.

[33] E. Eldele et al., "Time-series representation learning via temporal and contextual contrasting," 2021, *arXiv:2106.14112*.

[34] D. Lemire and M. E. O'Neill, "Xorshift1024*, xorshift1024+, xorshift128+ and xoroshiro128+ fail statistical tests for linearity," *J. Comput. Appl. Math.*, vol. 350, pp. 139–142, Apr. 2019.

[35] D. Liang, A. Liu, L. Wu, C. Li, R. Qian, and X. Chen, "Privacy-preserving multi-source semi-supervised domain adaptation for seizure prediction," *Cognit. Neurodynamics*, vol. 18, no. 6, pp. 3521–3534, Dec. 2024.

[36] N. Qi, Y. Piao, Q. Wang, X. Li, and Y. Wang, "Semi-supervised seizure prediction based on deep pairwise representation alignment of epileptic EEG signals," *IEEE Access*, vol. 12, pp. 119056–119071, 2024.

[37] A. L. Goldberger et al., "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, Jun. 2000.

[38] S. Zhao et al., "A 0.99-to-4.38 uJ/class event-driven hybrid neural network processor for full-spectrum neural signal analyses," *IEEE Trans. Biomed. Circuits Syst.*, vol. 17, no. 3, pp. 598–609, Jun. 2023.